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Chapter 7 Charles B. Fleddermann Slide.pdf
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Here is the organized text extracted from the PDF:

Chapter 7: Ethical Issues in Engineering Practice

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Engineering Ethics (CHARLES B. FLEDDERMANN)

Objectives:

- **Determine** what ethical issues arise in engineering practice with regard to the environment.
- **Decide** how engineering practice is impacted by computer technology.
- **Learn** about ethical issues that arise in the course of research.

Ethical Issues in Engineering Practice:

There are many unique ethical issues that arise in engineering practice that may not be encountered in other professions:

- Environmental Ethics
- Computer Ethics
- Ethics and Research

Environmental Ethics:

One of the most important political issues of the late 20th and early 21st centuries has been **environmental protection** and the rise of the **environmental movement**. This movement has sought to control the introduction of toxic and unnatural substances into the environment, protect the integrity of the biosphere, and ensure a healthy environment for humans.

Engineer's Responsibility to Protect the Environment:

- Engineers have a responsibility to protect the environment by:
 - Finding solutions to the problems caused by modern technology.
 - Using their knowledge and skills to help protect the environment.

This duty is also spelled out in many engineering codes of ethics.

Concepts:

- Engineer's responsibility for the environment is denoted by terms like **"sustainable design"** or **"green engineering"**.
- These concepts incorporate ideas to ensure that designs do not harm the environment.

Characteristics of Sustainable Design:

1. Ensures that a product has minimal environmental impact during its use.
 2. The product can be manufactured and disposed of without harming the natural world.
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The Moral Understanding of the Environment:

Questions Hard to Answer

To explore the environment's moral status, we consider some fundamental questions:

- Do we belong to nature, or does nature belong to us?
- If animals can suffer and feel pain like humans, should they have moral standing?

- If animals have moral standing, how far does this extend to other life forms like trees?

The Western Ethical Tradition vs. The Environmental Movement:

1. **Western Ethical Tradition:** Only human beings have moral understanding. Animals and plants are important only in respect to their usefulness to humans.
 2. **The Environmental Movement:** Humans are just one component of the environment, and all components have equal standing.
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Approaches to Resolving Environmental Problems:

1. Cost-Benefit Analysis:

- Derived from **utilitarianism**, it means analyzing a problem based on the benefits of reducing pollution and improving human health against the costs required to solve the problem.

Problems:

- Human lives may be deemed costless.
- Difficult to assess costs and benefits accurately.
- Does not necessarily consider who shoulders the costs and who gets the benefits.

2. Cost-Oblivious Approach:

- The cost is not important; instead, we focus on making the environment as clean as possible.

Problems:

- Difficult to uphold and enforce in a modern society.
 - Defining “as clean as” is challenging to agree on.
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Responsibilities of Engineers to the Environment:

- **Professional ethics:** Hold safety of people and the environment to be of paramount importance.
- **Personal ethics:** Determine the best course when confronted with an environmental problem.

An engineer should not be compelled by their employer to work on a project that they find ethically troubling, including projects with severe environmental impacts.

Key Principles:

1. Engineers must make decisions only in areas in which they are competent.
 2. Engineers should seek advice from experts like biologists, public health experts, and physicians to understand the environmental consequences of a project.
 3. Engineers must follow applicable federal and state regulations, ensuring their work is conducted in an environmentally safe manner.
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Computer Ethics:

- **Privacy**
- **Hacking**
- **Copyright infringement**

Computers have become an essential tool in engineering and business, bringing both benefits and serious ethical issues.

Engineers' Role with Computers:

The engineer's roles as **designer, manager, and user** of computers bring with them a responsibility to foster the ethical use of computers.

Computers as a Tool for Unethical Behavior:

Many uses of computers merely extend unethical acts. For example, they can be used to more efficiently steal money from a bank.

Unfortunately, technology to detect and prevent this crime lags behind the technology available to commit it.

Privacy:

- **Privacy** refers to the basic right of an individual to control access to and use of information about themselves.

Why is privacy an ethical issue?

- Invasions of privacy can lead to harm through harassment or blackmail.
- Unauthorized use of personal information is theft.

Impact of Computers on Privacy:

- Computerization makes retrieval of files easier, reducing storage space but also making unauthorized access easier.

Hacking:

Hacking involves unauthorized access to databases, implanting false information, altering existing information, or spreading viruses. This violates privacy rights, even if the hacker does not use the information.

Copyright Infringement:

Computers make it easy to share copyrighted material illegally. This diminishes creators' ability to profit from their work, which ultimately discourages creative activity.

Computers as an Engineering Tool:

Computers are essential tools in engineering, used for:

- Writing documents
- Financial calculations
- Keeping records
- Scheduling software
- Developing project plans

Unique to engineering, computers are also used for:

- **Design tools**
 - **Components integrated into engineered systems**
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Computer Design Tools:

Numerous software packages are available for the design of engineered devices and structures, making the work more efficient and removing some tedious aspects. However, these tools also lead to ethical issues.

Ethical Questions About Computer Design Tools:

- Who is responsible when a software flaw leads to the failure of a bridge?
- Who is at fault when software is used for a problem it is not suited for?
- What happens when existing software is applied to new and innovative designs?

The answer is clear: Software can never substitute for **good engineering judgment**. The engineer using the software is still responsible for the design and testing.

Tips for Engineers When Designing:

- Ensure the software is appropriate for the problem.
 - Stay updated on software flaws and use the latest version.
 - Verify the results of computer-generated designs.
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Integration of Computers into Engineered Systems:

Computers are part of many engineered systems, such as those in modern automobiles. These systems control functions like emissions and braking, improving efficiency and safety.

However, when designing systems with embedded computers, engineers must:

- Ensure that software is adequately tested.
 - Ensure that humans can intervene when necessary.
 - Ensure hardware redundancy for safety systems, not relying solely on software.
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Autonomous Computers:

Autonomy refers to the ability of computers to make decisions without human intervention.

Negative Implications of Autonomy:

- Some autonomous systems, like those in **stock markets** or **military weapons**, can cause instability if they malfunction.

Applications of Autonomy:

- **Manufacturing:** Autonomous computers can be useful for monitoring processes.
- **Stock Trading:** Automated systems can create unstable situations, as seen in the 1987 market crash.
- **Weapons Systems:** Autonomous weapons could lead to unintended conflict due to malfunctions or misinterpretations.

While autonomy can increase productivity and efficiency, human control is necessary to prevent disasters.

Computer Code of Ethics:

Organizations have developed codes of ethics for computer use. These guidelines should not replace sound moral reasoning and judgment.

Ethical Issues in Engineering:

Examples:

1. Accidental overdoses in medical radiation therapy systems (e.g., **Therac-25**).
2. Legal cases like **Avanti Corp. vs. Cadence Design Systems**.

These cases raise significant ethical concerns regarding the use of computers in engineering.

This is a comprehensive breakdown of the document. Let me know if you need further adjustments or specific sections to be elaborated!